If the media on the two sides of a thin lens have same refractive index, then the two focal lengths are equal. We will mostly be using the secondary focus F_2 in our further discussions.

6.6.2 Sign conventions on focal length for lens

The sign conventions for thin lenses differ only in the signs followed for focal lengths.

(i) The sign of focal length is *not decided* on the direction of measurement of the focal length from the pole of the lens as they have two focal lengths, one to the left and another to the right.

(ii) The focal length of the thin lens is taken as positive for a converging lens and negative for a diverging lens.

The other sign conventions for object distance, image distance, radius of curvature, object height and image height (except for the focal lengths as mentioned above) remain the same for thin lenses as that of spherical mirrors.

6.6.3 Lens maker's formula and lens equation

Let us consider a thin lens made up of a medium of refractive index n_2 placed in a medium of refractive index n_1. Let R_1 and R_2 be the radii of curvature of two spherical surfaces ① and ② respectively and P be the pole as shown in Figure 6.34. Consider a point object O on the principal axis. A paraxial ray from O which falls very close to P, after refraction at the surface ① forms image at I'. Before it does so, it is again refracted by the surface ②. Therefore, the final image is formed at I.

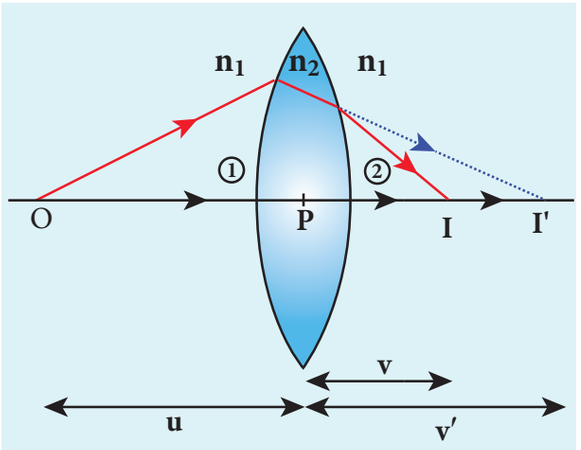


Figure 6.34 Refraction through thin lens

The general equation for the refraction at a single spherical surface is given by the equation (6.56) is,

$$\frac{n_2}{v}-\frac{n_1}{u}=\frac{(n_2-n_1)}{R}$$

For the refracting surface ①, the light goes from n_1 to n_2.

$$\frac{n_2}{v'}-\frac{n_1}{u}=\frac{(n_2-n_1)}{R_1} \qquad (6.58)$$

For the refracting surface ②, the light goes from medium n_2 to n_1.

$$\frac{n_1}{v}-\frac{n_2}{v'}=\frac{(n_1-n_2)}{R_2} \qquad (6.59)$$

For surface ②, I' acts as virtual object.

Adding the above two equations (6.58) and (6.59)

$$\frac{n_1}{v}-\frac{n_1}{u}=(n_2-n_1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

On further simplifying and rearranging,

$$\frac{1}{v}-\frac{1}{u}=\left(\frac{n_2-n_1}{n_1}\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

$$\frac{1}{v}-\frac{1}{u}=\left(\frac{n_2}{n_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right) \qquad (6.60)$$

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